H · Joining and positioning the countertops

Two people. Both tops are face down with legs, feet and rails fitted, every OLOV pre-set to 802 (80.2 cm), every pad set with the foot height gauge (102); the ALEX units stand in place with their rings on.

Parts	Hardware	Tools
P01 top A · P02 top B · P03 ALEX 1 · P04 ALEX 2 · P05 OLOV leg · P06 rail A · P07 rail B · P08 strut fitting × 2 · P09 foot · P10 ring · P11 mending plate × 2 (joining)	H03 #10 × 1 in × 2 (rail B holes 1, 2) · H04 #10 × 5/8 in × 8 · H05 #10 washer × 8 · H08 3/8-16 × 1-1/2 in bolt × 6 · H09 3/8 washer × 12 · H10 3/8-16 nylon nut × 6	Drill guide (SET tower, pucks 25 RAIL and 12 PLATE), 1/8 in bit, Robertson #2 on a 3 in extension, level, straightedge, tape, torque wrench or 15 cm wrench

1

Stand top A. With a helper, turn top A over and stand it on its legs: lower the ALEX end first so the four pads drop into their rings on ALEX 1, then the leg end. Level along x and along y by turning the OLOV feet and the round feet until it reads flat and all four pads bear with no rock. Target: surface 830 (83.0 cm) at every corner. SOURCE-DEFINED

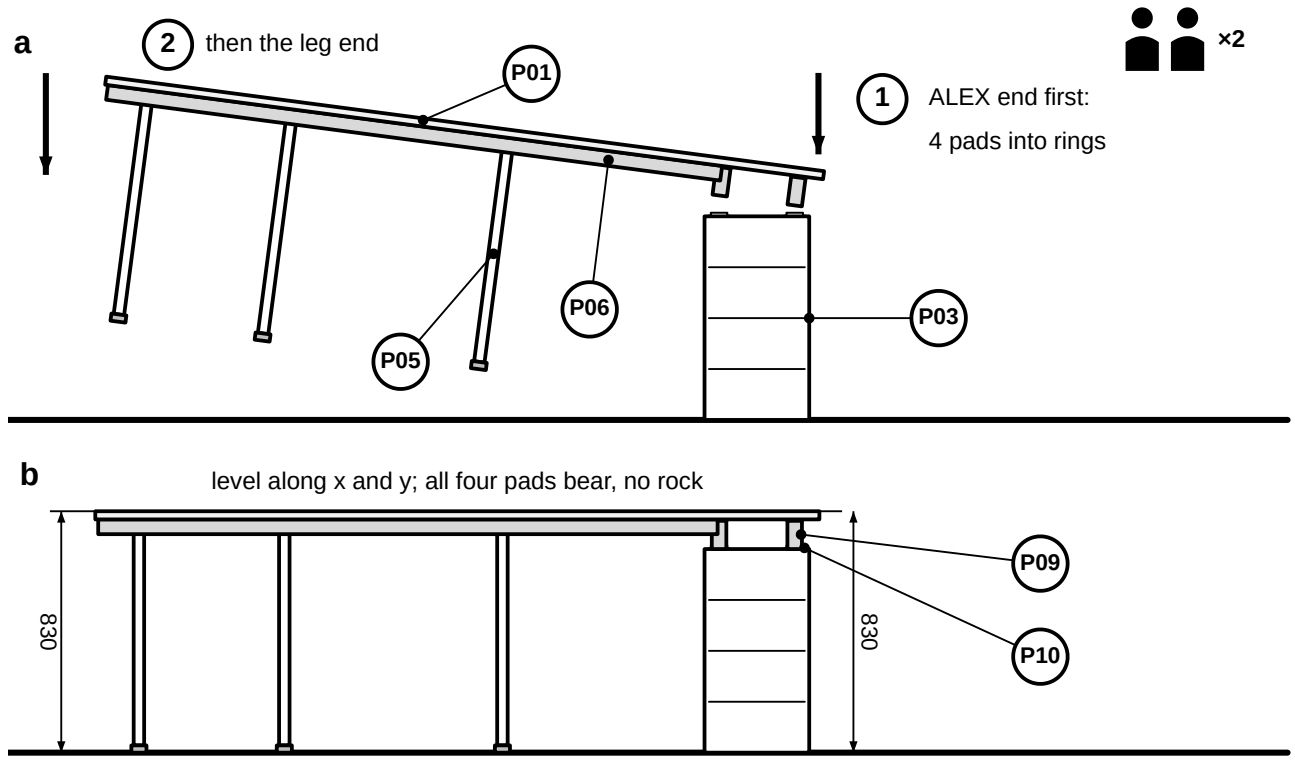


Fig. H-1 · Top A from the front (1:26). a: ALEX end first, the pads drop into the rings; b: leg end down, level, 830 at both ends. Rail A hangs under the top; it never touches ALEX 1.

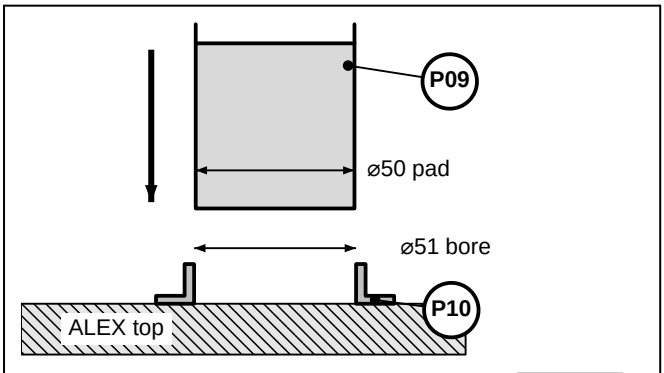


Fig. H-2 · Pad into ring (section, 1:2.4). Bore 51 VERIFIED; pad 50 MEASURE ON HARDWARE

Once the pads are in the rings, top A cannot be pushed.
If the pads miss, or the top is not 30 (3 cm) off a wall, lift the ALEX end a few millimetres and slide the ALEX unit, not the top. The wall gauge's 30 leg, or a 3 cm offcut, against the wall is the easiest gauge.

830 assumes ALEX units 700 tall ASSUMPTION / FIELD CHECK until measured. With the optional ALEX feet the surface rises by their height (20, 40 or 60).

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2

Stand top B and slide it home.

1. Wind top B's OLOV legs 2–3 mm short, so rail B's corner end (it projects 120 past B's end) can slide under top A.
2. Turn top B over onto its legs with its pads in the rings on ALEX 2.
3. Push top B (ALEX 2 slides with it) until rail B's corner end touches the outer face of rail A's vertical leg and B's end meets A's front edge at y 650 (65 cm), 30 off the side wall.
4. Wind B's legs up until the two surfaces are flush.

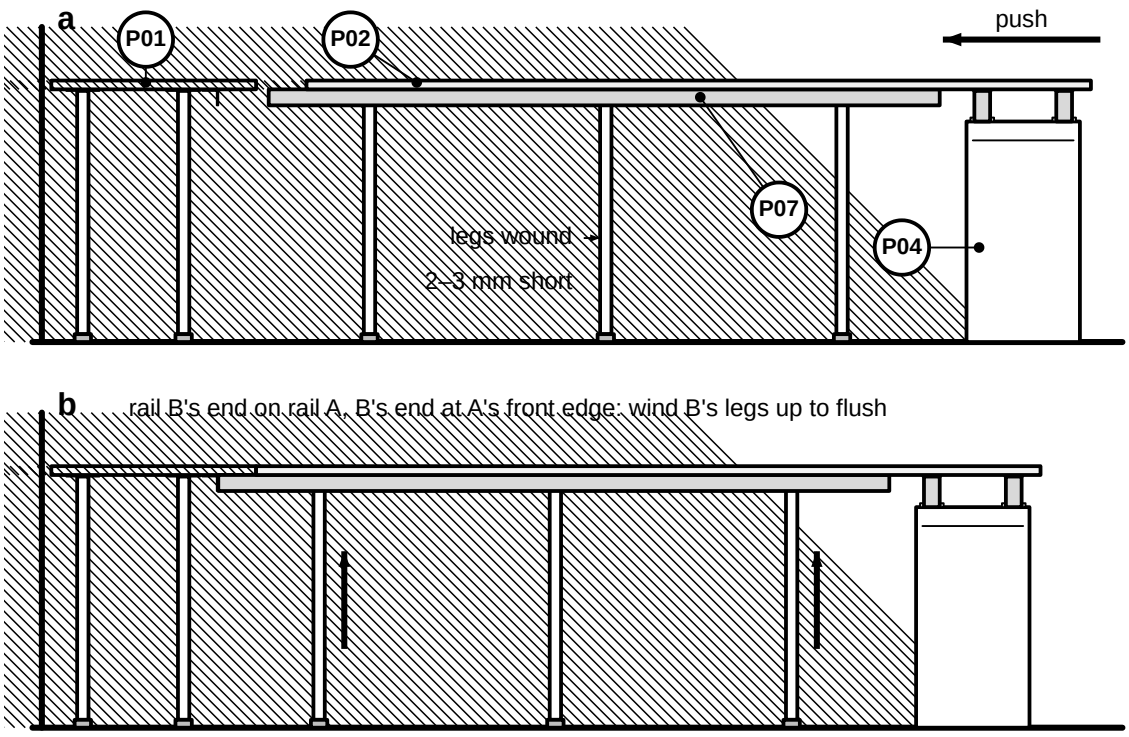


Fig. H-3 · Top B seen from the knee space, looking at the side wall (1:24); top A cut beyond B's inside edge, ALEX 1 not shown. a: legs short, push; b: home, wind the legs up. The rails rest only on the tops; neither touches an ALEX.

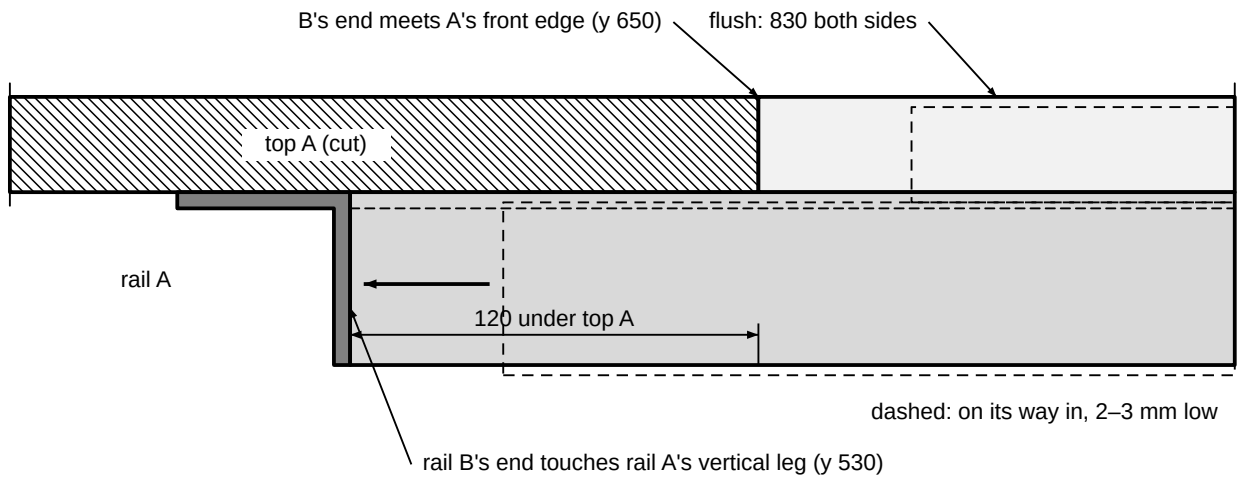


Fig. H-4 · The joint in section, same view (1:2.2). Dashed: top B on its way in, 2–3 mm low. Solid: home. Rail B's flat leg bears on top A's underside for its first 120; its holes 1 and 2 are drilled in step 5.

!

Never force the corner.
If the corner will not meet, or the surfaces will not come flush, adjust leg height, not the joint. Do not hammer, clamp or lever the tops together.

3 **Check under the corner, then level the joint.** Rail B's end must touch the face of rail A's vertical leg with no gap, and rail B's outer face must line up with the 52.0 cm (520) mark on rail A. If not, lift top B a little and move it. Then bring the two surfaces level at the joint by turning the OLOV feet and the round feet, never by packing anything under the joining plates. A fingertip across the joint finds no step.

4 **Joining plates P1 and P2, from below.** Screw each plate on the outline you drew: y 533–787 (53.3–78.7 cm), long axis along y, centred x 150 and 350. They are not centred on the joint: each starts 3 clear of rail A. Pilot every hole 1/8 in, 12 deep, with the drill guide's **12 PLATE** puck held up overhead; drive H04 #10 × 5/8 in with an H05 washer. Two screws in each top. SOURCE-DEFINED

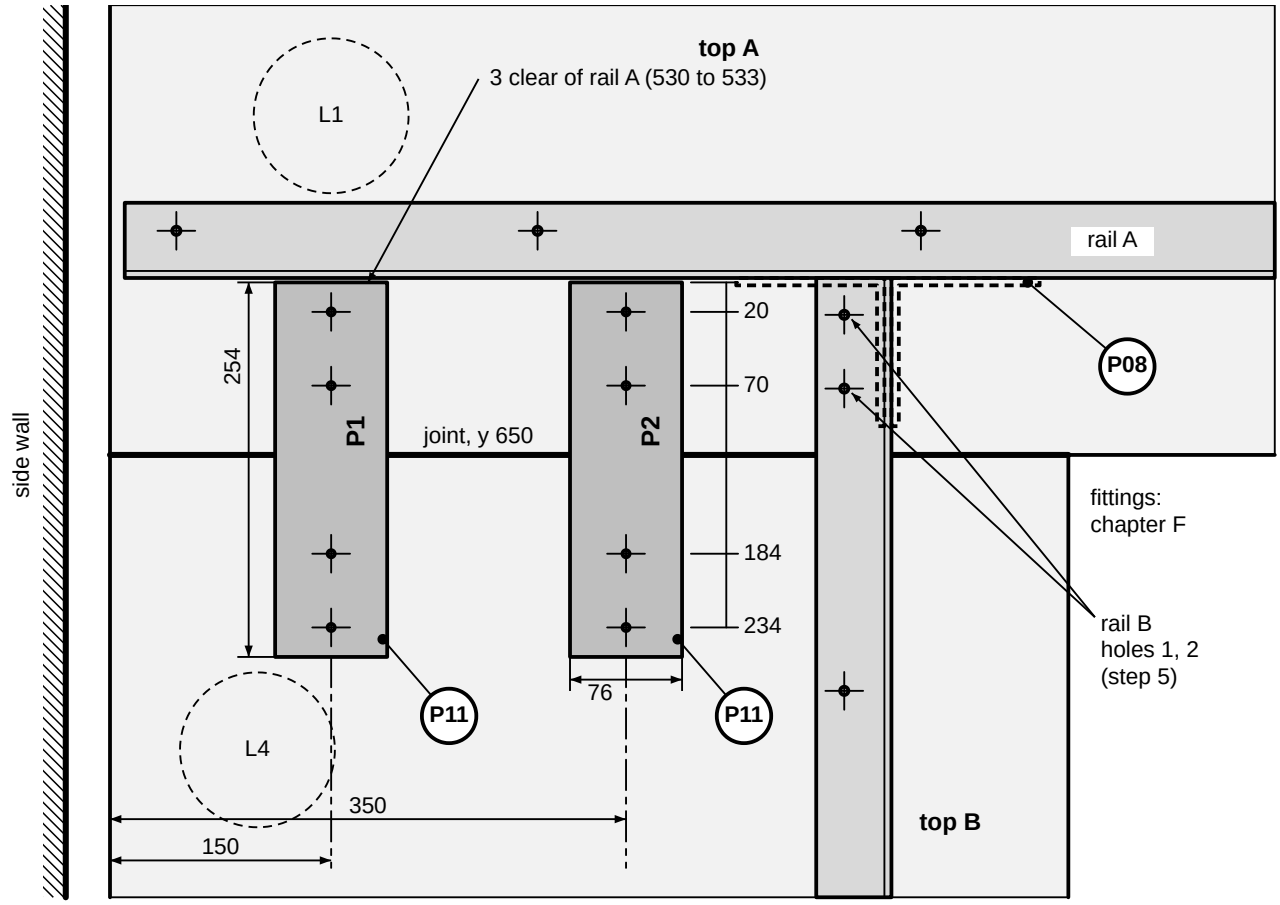


Fig. H-5 · The corner from above, looking through the tops (1:5). Datum: top A's back-left corner, x along the back wall, y along the side wall. Plate holes at 20, 70, 184 and 234 from the plate end nearest rail A. Dashed: OLOV plates L1 and L4, and the two fittings (orientation in chapter F).

Plate	Centre x	Span y	Screws in top A, y	Screws in top B, y	Pilot
P1	150	533–787	553, 603	717, 767	⌀3.2 (1/8 in) × 12, puck 12 PLATE
P2	350	533–787	553, 603	717, 767	same

The plates align the tops; they do not carry the corner. Every plate hole is more than 40 (4 cm) from an edge, so all four get a screw.

5 **Rail B meets top A.** Through rail B's holes 1 and 2 (25 and 75 from its corner end, both under top A) drill a 1/8 in pilot with the drill guide's **25 RAIL** puck on the flat leg: 4.8 of steel plus 20 into the wood. Drive H03 #10 × 1 in with the Robertson #2 bit on the 3 in extension until the head seats. SOURCE-DEFINED

6

Bolt on the two fittings, then tighten. Orientation is in chapter F: outside fitting heel at the outside corner, inside fitting heel at the inside corner with its gusset pointing at the floor.

1. Rail A: four H08 bolts from the fitting side, H09 washer under each head, washer and H10 nylon-insert nut on the inside of the leg. Finger-tight.
2. Rail B: two H08 bolts through outside fitting, leg and inside fitting, washer each end, H10 nut. Finger-tight.
3. Only when the surfaces are flush (step 3) and rail B's end touches rail A: tighten all six to **25 N·m (18 ft·lb)**; without a torque wrench, one hand's full pull on a 15 cm wrench, then stop. SOURCE-DEFINED

Tight, not snug: the fittings' 21 × 12 slots leave 2.5 of play across and 11.5 along each bolt, and the 7/16 in rail holes 1.6, so only the clamping force holds the corner. Fitting legs 100.5 × 41, 5 thick; slot centres 29.15 and 78.45 from the heel VERIFIED. Rail bolt holes: rail A 549.2 and 598.5 (outside fitting) and 485.9 and 436.6 (inside fitting) from the wall end; rail B 29.2 and 78.5 from the corner end VERIFIED.

7

Checks. Do every one; if one fails, fix it with the feet or the legs, never at the joint.

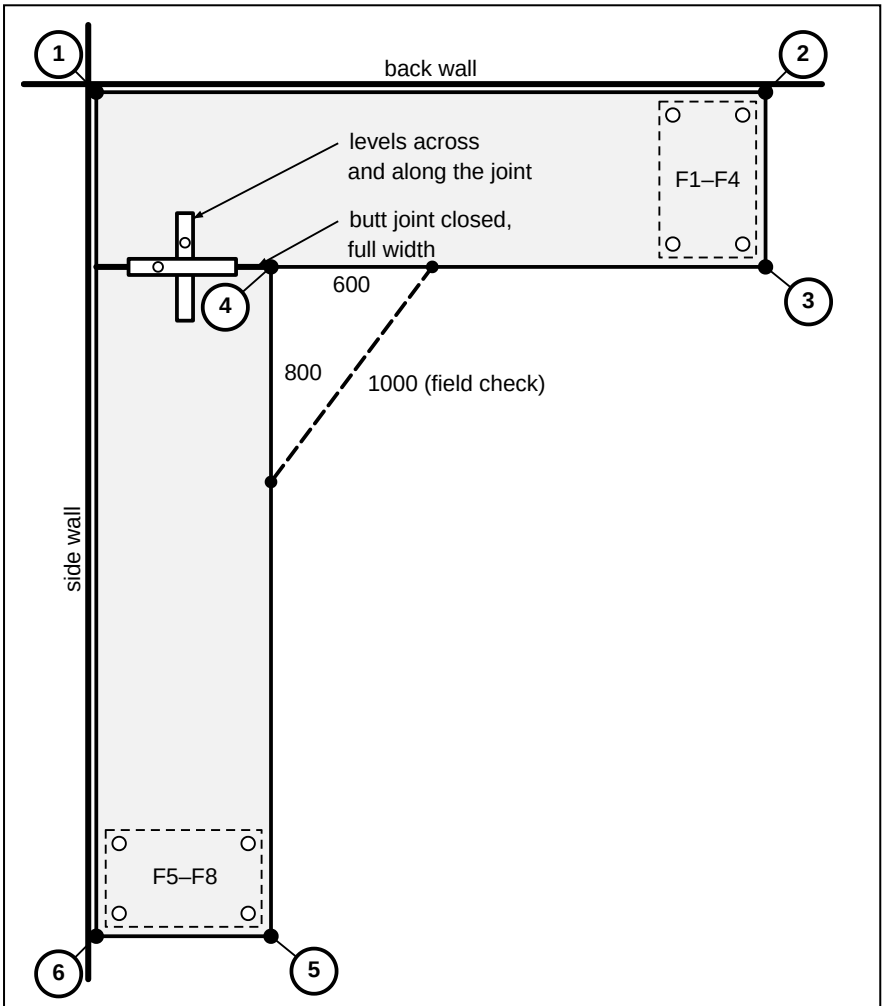


Fig. H-6 · The checks in plan (1:28). Datum: top A's back-left corner.

- **Square:** the butt joint is closed across its full 650 width (no light), rail B's end touches rail A, rail B's outer face is on rail A's 520 mark.
- **Level:** a level across the joint and along it reads flat both ways; a fingertip finds no step.
- **Height:** surface 830 ± 3 at all six outer corners, 1–6. SOURCE-DEFINED
- **Pads:** all eight bear. Turn each of F1–F8 by hand in its ring: none spins free or rocks.

! **FIELD CHECK, not in the build manual** ASSUMPTION / FIELD CHECK

Optional 3-4-5 triangle at the inside corner: mark 600 along A's front edge and 800 along B's inside edge; the diagonal between the marks is 1000 when the L is square. Write down what it reads. If it is off, check the butt joint for a gap first, then move ALEX 2 with top B on it. Never force the corner.

The corner load test (20 kg on top B's inside corner), the sideways push and the settling record are in the final checks. Never stand or sit on the corner.

8b · Wall-gap trim (optional)

> **Do this after chapter I.**
> The clamp pieces wrap round the VIVO clamp backs, so both clamps must already be fitted and tight. The trim goes on before the final push to the walls.

Parts P19 (corner, two 80 fillers, two clamp pieces, 240 segments, connector tiles), H07 #4 × 1/2 in flat head, the drill guide's **10 EDGE** puck with a 5/64 in bit, the wall gauge.

8 **Measure, set, print.** With the desk in place against both walls, measure the gap between each top's raw back edge and the wall at five or six points along each wall. Set `gap` in `parts/accessories/wall-gap-common.scad` to the smallest value minus 1 and print the trim. MEASURE ON HARDWARE (design gap 30, so `gap` 29 as saved).

Wall	From the wall corner outward (desk positions)	Slotted segments
back	corner (to 200), 80 filler (200–280), clamp piece (280–520), 8 × 240, 1 × 50 (to 2490)	3rd (1000–1240), 8th (2200–2440)
side	corner (to 200), 80 filler (200–280), clamp piece (280–520), 10 × 240, 1 × 220 (to 3140)	3rd (1000–1240)

The corner piece covers each wall's first 200 of desk. 24 connector tiles: 11 joints on the back wall, 13 on the side wall. Cut list from parts/README.md.

9 **Pull the desk out, 400 (40 cm) from both walls.** Two people, one at each ALEX unit: slide the whole desk away from both walls. The ALEX units slide on their feet and the pads stay in their rings. Nothing comes apart.

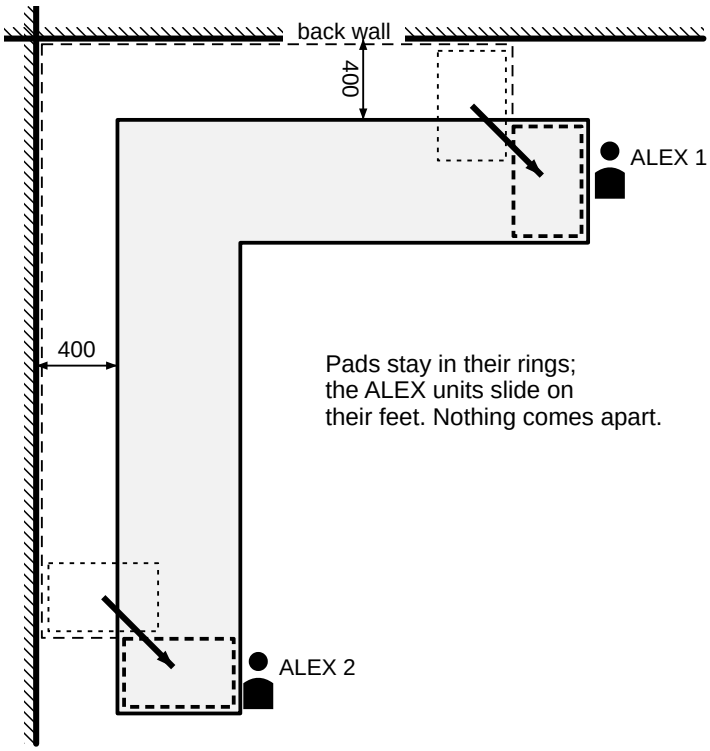
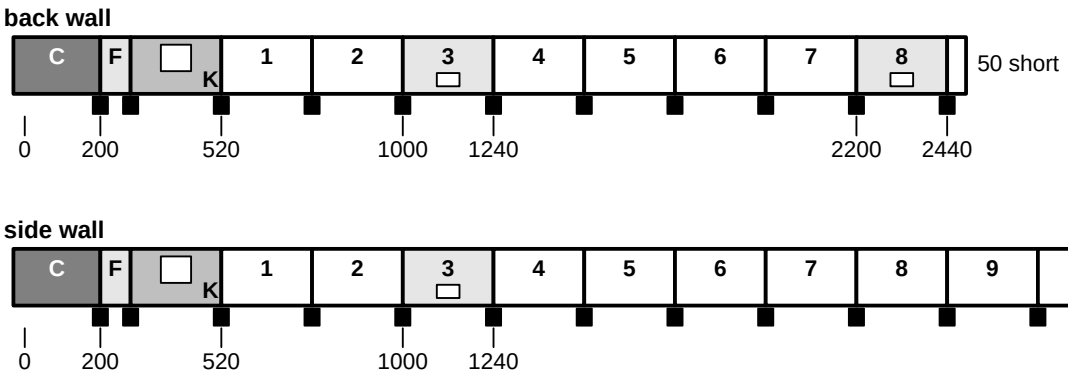


Fig. H-7 · Plan (1:40). Dashed: the desk in place, 30 off both walls. Grey: pulled 400 off both, so the raw edges are open to work on.

10 **Build each chain from the corner outward.** Corner piece, 80 filler, clamp piece with its notch round the clamp back, then the 240 segments (slotted ones where the cut list says), ending with the short piece. Butt every joint and push a connector tile up into the pockets.



C corner · F 80 filler · K clamp piece (white: notch round the clamp back)
1, 2 … 240 segments, grey with a slot = slotted · ■ connector tile under every joint

Fig. H-8 · Chain order along both walls, desk positions in mm from the corner (1:20). The notch (white, in K) wraps the clamp back, 80 long and 17.5 proud VERIFIED.

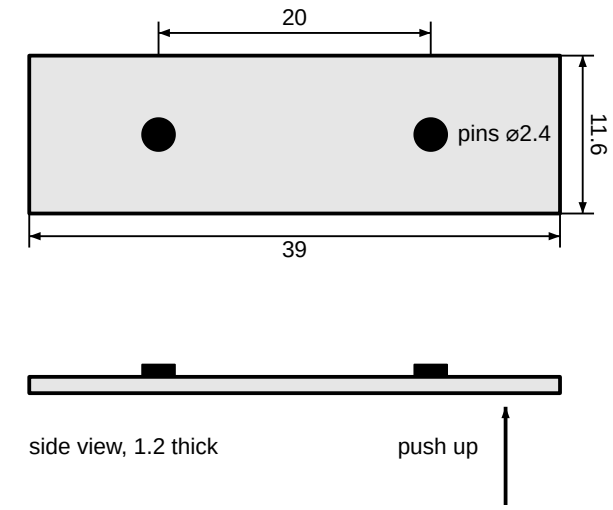


Fig. H-9 · Connector tile (1.8:1), one under every joint, pins up. It bridges the 20 × 12 pockets of two butted pieces.

Every piece, the corner included, has a pocket on its underside at each end, over the gap. Butt two pieces, then push a tile up into both pockets until its pins seat: the joint cannot step, bow or drift, and nothing shows on top.

If a wall is out and the notch misses the clamp, change the 80 filler's `seg` and reprint it, nothing else.

11 **Pilot and screw.** Through each lip hole, pilot 5/64 in, 10 deep, drill horizontal, with the drill guide's **10 EDGE** puck held against the raw edge. Drive an H07 #4 × 1/2 in flat head into each: two per segment, one per corner leg, one each side of a clamp notch.

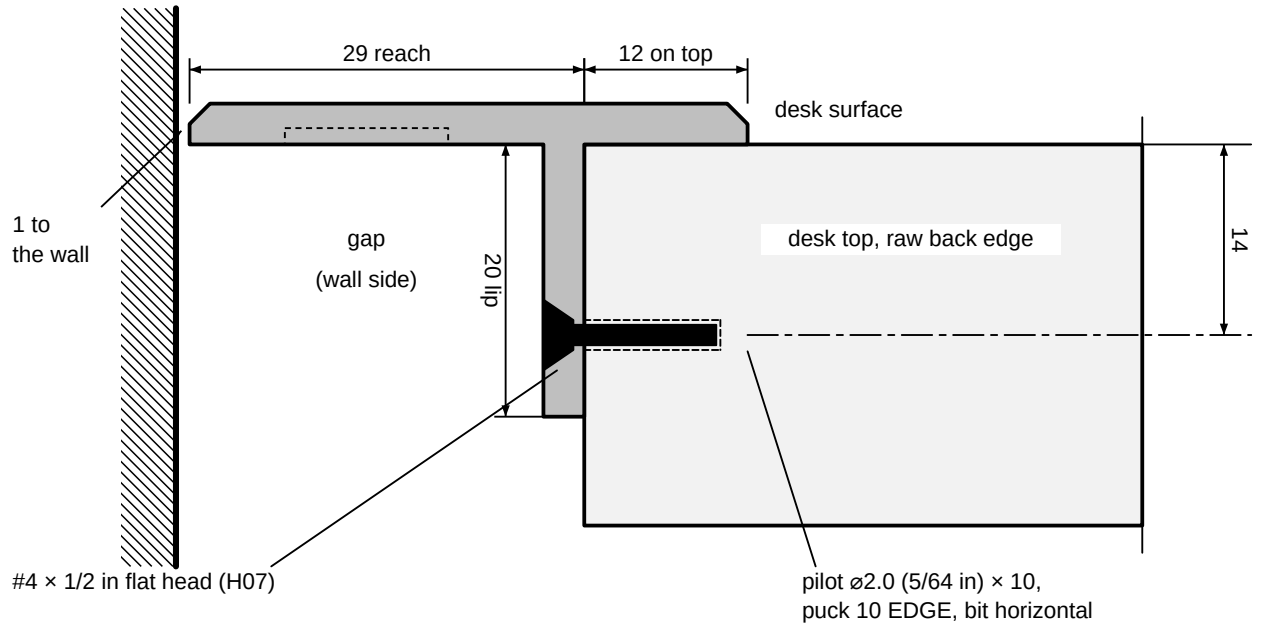


Fig. H-10 · Trim on the raw back edge, section (1.8:1). 12 rests on the surface, the 20 lip drops into the gap, the screw sits 14 below the surface: nothing shows on top and nothing goes into the top face. Profile from parts/accessories/wall-gap-common.scad SOURCE-DEFINED; top 28 thick MEASURE ON HARDWARE.

12 **Slide the desk back.** Two people again, one at each ALEX, until the wall gauge's 65 leg just fits between the wall and each ALEX. The trim's outer edge stops 1 short of the wall.

Retighten every screw after two weeks.

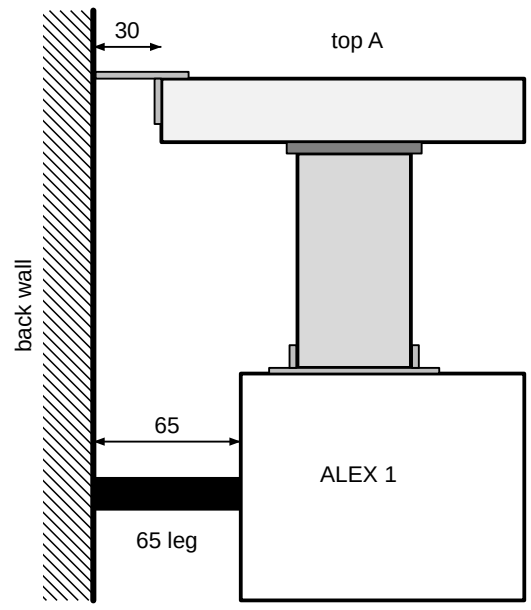


Fig. H-11 · Wall gauge at ALEX 1, section along the back wall (1:3.3).